

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2023

Course: B. Tech. Branch: Civil Engineering Semester: VI

Subject Code & Name: BTCVC403 Structural Mechanics - I

Max Marks: 60

Date: 18/07/2023

Time: 2:00 to 5:00 PM

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve <u>any two</u> of the following.		12
A) A simply supported beam of span 4 m is carrying a UDL of 2 kN/m over the entire span. Find the maximum slope and deflection of the beam. Take EI for the beam as $80 \times 10^9 \text{ N-mm}^2$.	CO2	06
B) A cantilever of length '2a' is carrying a load of 'W' at the free end and another load of 'W' at its centre. Determine the slope and deflection of cantilever at free end.	CO2	06
C) Enlist the different methods to find deflection in beams and explain in detail double integration method?	CO2	06
Q.2 Solve <u>any two</u> following questions.		12
A) Define the terms: a) Strain Energy, b) Strain Energy Density and c) Castigliano's theorems	CO2	06
B) Use strain energy method & determine the deflection at the free end of a cantilever beam of length 'L' subjected to point load P at the free end.	CO2	06
C) A simply supported beam of span 6 m & M.I. as $2I_0$ with one side as overhanging of span 2 m with M.I. as I_0 . Beam is subjected to UDL of 45 kN/m throughout the span. Determine the deflection of free end of overhanging beam. Use Unit load method.	CO2	06
Q. 3 Solve <u>any two</u> of the following.		12
A) Determine static indeterminacy & elastic curve of below beams: a) Simply supported beam subjected to UDL throughout the span. b) Cantilever beam subjected to point load at free end. c) Fixed beam subjected central point load.	CO1	06
B) Determine the reaction components of propped cantilever of span L subjected to uniformly distributed load w/unit length.	CO1	06
C) A fixed beam of span 'L' subjected to UDL 'w' kN/m throughout the span. Find SFD and BMD. Locate point of contraflexure. Also find the maximum deflection. Flexural rigidity 'EI' is constant.	CO1	06

Q.4 Solve <u>any two</u> of the following.		12
A) State and explain theorem of three moments (Clapeyron's theorem) for analysis of two span continuous beam simply supported ends. Each span of length L , M.I. as I , EI as flexural rigidity.	CO4	06
B) Two span continuous beam ABC of span AB 2 m & span BC 2 m. Span AB is subjected to UDL of 50 kN/m. Span BC is subjected 40 kN at distance 0.5 m from C. Draw SFD and BMD.	CO4	06
C) Define & explain the following terms: A) Carry over moment B) Carry over factor C) Stiffness factor	CO4	06
Q. 5 Solve <u>any two</u> of the following.		12
A) State and explain the assumptions made in slope deflection method (SDM) while developing this method.	CO3	06
B) Analyse the two-span continuous beam ABC by slope deflection method having both ends as fixed one. Span AB is having 4 m length, M.I. as ' I ' & subjected central point load of 40 kN. Span BC is having 6 m length, M.I. as ' $2I$ ' & subjected to UDL of 20 kN/m over entire span BC. Draw SFD & BMD.	CO3	06
C) Analyse the two-span continuous beam ABC by slope deflection method having one end as fixed one & other end as roller support. Span AB is having 6 m length, M.I. as ' I ' & subjected point load of 60 kN at distance of 4m from fixed support A. Span BC is having 6 m length, M.I. as ' I ' & subjected to UDL of 30 kN/m over entire span BC. Draw BMD.	CO3	06

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